

Master Technical Architecture

Offline-first casualty-care PWA, conflict-aware sync backend, and provincial-EHR integration (Ontario Health).

PROTOTYPE — NOT FOR CLINICAL USE

Repository: **akromi/Emergency-Medical-Support-System**

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TRIAGE-LINK — Medical Device Certification Roadmap

Purpose. A phased plan to take TRIAGE-LINK from an offline-first **prototype** to a **commercial, certified Software as a Medical Device (SaMD)** — covering the regulatory strategy, quality management system, standards-compliant engineering, clinical evaluation, certification audits, and post-market obligations.

This complements [docs/ROADMAP.md](#) (the engineering build plan: prototype → production system). That roadmap delivers the *capabilities*; this one wraps them in the *regulated process and evidence* a certified medical device requires. Cross-references are noted as (*build: Phase N*).

⚠ Not legal/regulatory advice. Classification and pathways are product-, claim-, and market-specific. Engage a regulatory affairs professional, an ISO 13485 registrar, and (for EU) a Notified Body **early** — these decisions shape cost and timeline more than any code.

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■ 1. The one decision that drives everything

Is TRIAGE-LINK a medical device? A SaMD is software intended for a *medical purpose* (diagnosis, treatment, prevention, monitoring). Two paths from today's prototype:

If the intended use is...	Then...
A documentation / record-keeping tool (captures and transmits what a clinician decides)	May fall outside SaMD scope — far lighter regulatory burden. Most defensible commercial first step.
Software that drives or informs clinical decisions (e.g. computes triage category, burn TBSA, or flags acuity that clinicians act on)	Is SaMD — full design controls, ISO 14971, IEC 62304, clinical evaluation, certification.

Today's prototype already computes acuity-adjacent outputs (Lund–Browder burn **TBSA**, START triage tagging). If those outputs are *presented as clinical guidance*, the product trends toward SaMD. **Define the intended use statement precisely** — it determines classification, and classification determines everything below. This roadmap assumes the **SaMD** path (the higher bar); de-scope if the documentation-tool path is chosen.

Action: write a one-paragraph **Intended Use / Indications for Use** statement and an **intended-user / intended-environment** description, and record them as the controlling inputs. Everything downstream traces back to these.

■ 2. Intended use & device classification

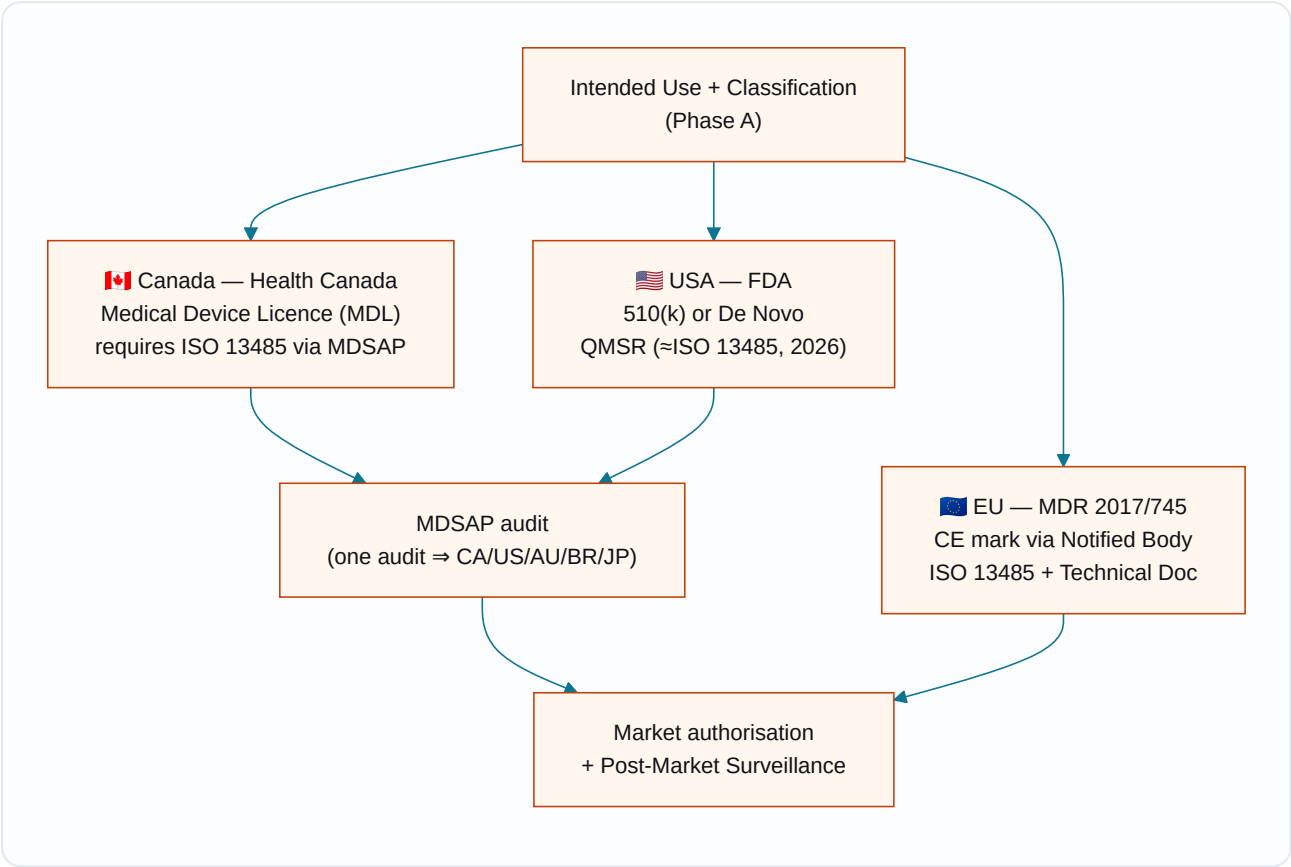
Classification follows the **risk** posed by the software's role: the significance of the information to a healthcare decision × the state of the healthcare situation (IMDRF SaMD framework). Likely landing zones (to be confirmed by regulatory assessment):

Framework	Likely class	Rationale
IMDRF SaMD	Category II–III	"Drives/informs clinical management" in a "serious/critical" (emergency) situation pushes the category up
Health Canada (SOR/98-282)	Class II (possibly III)	Software informing triage/treatment in emergencies
US FDA	Class II (510(k) or De Novo if no predicate)	Most clinical-decision-support SaMD; some CDS is exempt under the 21st Century Cures Act criteria
EU MDR 2017/745	Class IIa/IIb (Rule 11)	MDR Rule 11 classifies most clinical software ≥ IIa; "serious deterioration" → IIb

Emergency / mass-casualty context raises the stakes: a wrong or lost record can affect a critical decision, which is exactly what pushes class and evidence requirements upward. Plan for **Class II at minimum**.

3. Target markets & regulatory pathways

Pick the **launch market first** (sequence the rest) — each adds a parallel submission track. Given the Ontario / Ontario Health focus, **Canada is the natural first market**.



Market	Authorisation	QMS basis	Notes
Canada	Medical Device Licence (Class II+)	ISO 13485 via MDSAP (mandatory)	MDSAP single audit also satisfies US/AU/BR/JP
USA	510(k) clearance, or De Novo if novel	QMSR (harmonises with ISO 13485, effective 2026)	Identify a predicate device early; De Novo if none
EU/UK	CE mark (EU MDR) / UKCA	ISO 13485 + MDR Technical Documentation	Notified Body capacity is a real schedule risk

Leverage MDSAP: a single MDSAP audit covers Canada, USA, Australia, Brazil, and Japan — the most efficient multi-market QMS strategy.

4. The standards stack

A certified SaMD is built **to** these standards, with evidence generated **as you go** (not retrofitted). The core set:

Standard	Scope	What it forces into the project
ISO 13485	Quality Management System	Document control, design controls, CAPA, supplier control, records
IEC 62304	Medical device software lifecycle	Software safety classification (A/B/C), planning, architecture, unit/integration/system V&V, SOUP/3rd-party management, problem resolution
ISO 14971	Risk management	Risk management file: hazard analysis, risk controls, benefit-risk, residual risk
IEC 62366-1	Usability engineering (human factors)	Use-specification, use-related risk, formative + summative usability validation
IEC 81001-5-1	Health software cybersecurity lifecycle	Secure SDLC, threat modelling, SBOM, vulnerability management
IEC 82304-1	Health software product safety	Whole-product requirements, validation, accompanying documentation
ISO/IEC 27001	Information security management	(Often paired) ISMS for the hosted backend
ISO 14155	Clinical investigation	If a clinical study is needed for evidence
ISO/TR 80002-2	Software validation	Validation of QMS/process software & tools

Privacy & health-data law (parallel, mandatory): PHIPA (Ontario), PIPEDA (Canada), HIPAA (US), GDPR (EU). Plus **HL7 FHIR conformance** to provincial profiles (already the product's integration boundary — an asset).

■ 5. Gap assessment — prototype vs. certified

What exists today vs. what certification requires. (Engineering items cross-ref [docs/ROADMAP.md](#).)

Area	Today (prototype)	Required for certification	Gap
Intended use / claims	Implicit ("not for clinical use")	Formal Intended Use, Indications, classification	★★★
QMS	None	ISO 13485 certified QMS	★★★
Design controls / DHF	Ad-hoc docs (good architecture docs exist)	Design History File: inputs → outputs → V&V → traceability	★★★
Risk management	Informal	ISO 14971 risk file, hazard analysis, benefit-risk	★★★
Software lifecycle	Modern TS, tests, CI	IEC 62304 plan, safety class, documented architecture, SOUP list, traceability	★★
Usability / HFE	Functional UI	IEC 62366 use-spec + summative validation with representative users	★★★
Cybersecurity	Fail-closed EHR selection, ATNA audit, idempotent writes, mTLS seam	IEC 81001-5-1 secure SDLC, threat model, SBOM, pen test, vuln mgmt	★★
AuthN/AuthZ	None on PWA	OIDC/SSO, MFA, RBAC, session management (<i>build: Phase 3</i>)	★★★
Encryption at rest	IndexedDB plaintext	Encrypted device store + encrypted backend/backups (<i>build: Phase 3</i>)	★★★
Audit trail	EHR access audited (ehr_audit , ATNA)	All-PHI CRUD audit, immutable/WORM, time-synced	★★
V&V evidence	~90 unit/integration tests	Documented, traced system-level V&V to requirements + risk controls	★★
Clinical evaluation	None	Clinical Evaluation Plan/Report; possibly clinical investigation	★★★
Labelling / IFU	README disclaimers	Instructions For Use, labelling, UDI	★★★
Post-market	None	PMS plan, vigilance, complaint handling, PSUR	★★★

Existing assets that accelerate certification: framework-free core (clean requirements/architecture traceability), strong test culture and CI, FHIR-standard interoperability, the architecture/deployment docs, fail-closed safety design, and per-access audit. These are real head starts — most prototypes have none.

■ 6. Phased certification roadmap

Seven phases. Phases **B–D** run with engineering in parallel; the key is that engineering output is produced **under** the QMS so it counts as evidence.

Phase A — Regulatory strategy & classification (months 0–3)

Objective: lock the controlling decisions before scaling spend.

- Author the **Intended Use / Indications for Use**, intended users, and use environment.
- **Classification assessment** (IMDRF + each target market); choose **launch market** & sequence.
- **Regulatory strategy** (pathway per market: MDL / 510(k)/De Novo / CE), predicate search (US).
- Assign **IEC 62304 software safety class** (A/B/C) — drives the depth of all software work.
- Engage a regulatory consultant; select ISO 13485 **registrar**/Notified Body (lead-time matters).
- **Deliverables:** Intended Use statement, Classification rationale, Regulatory Strategy, Project/Quality plan.
- **Exit gate:** classification & pathway signed off; safety class fixed.

Phase B — Quality Management System (ISO 13485) (months 2–8)

Objective: stand up the QMS that everything else is produced within.

- Establish the **Quality Manual**, SOPs (document/record control, design control, CAPA, supplier/SOUP control, risk management, software development, change control, complaint handling, post-market surveillance, release).
- Define org, roles, **Management Representative**; training records.
- Validate process/tool software (ISO/TR 80002-2) — issue tracker, CI, test tools.
- **Deliverables:** Quality Manual + SOP set, training records, validated toolchain.
- **Exit gate:** QMS operational; ready for internal audit. (*Stage-1 readiness.*)

Phase C — Design controls, risk & usability (months 4–12)

Objective: build the **Design History File** and risk/usability evidence.

- **Design inputs:** user needs → requirements (functional, performance, safety, security, regulatory, interoperability). Trace from the Intended Use.
- **Design outputs:** architecture, interface specs, the codebase — under change control.
- **ISO 14971 risk file:** hazard analysis (incl. emergency-context hazards: lost record, wrong-patient match, stale offline data, mis-sync), risk controls, benefit-risk, residual risk.
- **IEC 62366-1 usability:** use-specification, use-related risk analysis, **formative** studies, UI/IFU refinement; plan **summative** validation with representative responders.
- **IEC 62304 software file:** development plan, software architecture, detailed design, **SOUP list + SBOM**, unit/integration design, anomaly/problem-resolution process.
- **Traceability matrix:** user need ↔ requirement ↔ design ↔ risk control ↔ test.
- **Deliverables:** DHF (inputs/outputs/reviews), Risk Management File, Usability File, Software Dev File, traceability matrix.
- **Exit gate:** design reviews passed; requirements frozen for V&V.

Phase D — Engineering hardening to standard (months 6–14, parallel with build roadmap)

Objective: close the technical gaps as controlled design outputs.

- **Security (IEC 81001-5-1):** threat model, secure-coding standard, dependency/vuln management, secrets management, **penetration test**, security risk assessment.
- **AuthN/AuthZ:** OIDC/SSO + MFA, RBAC least-privilege (*build: Phase 3*).
- **Encryption:** device store at rest + backend/backups; TLS 1.3; mTLS to EHR (seam exists).
- **Audit:** extend ATNA audit to **all PHI CRUD**, immutable/WORM storage, time sync.
- **Reliability:** backup/restore, DR, the MCI-burst non-functionals (*build: Phase 5*).
- **Deliverables:** security risk assessment + pen-test report, SBOM, hardened build, updated DHF.
- **Exit gate:** security controls verified; no unmitigated high-severity findings.

Phase E — Verification, validation & clinical evaluation (months 12–18)

Objective: prove the device meets requirements and is clinically safe/effective.

- **Verification:** documented unit/integration/system tests **traced** to requirements & risk controls (build on existing ~90 tests, formalised under the QMS).
- **Validation:** the product meets user needs in the intended environment (incl. offline & MCI).
- **Summative usability validation** (IEC 62366) with representative users — pass criteria pre-defined.
- **Clinical Evaluation Plan/Report;** decide if a **clinical investigation** (ISO 14155) is needed for evidence; assemble real-world / literature evidence.
- **Deliverables:** V&V reports + traceability, Summative Usability Report, Clinical Evaluation Report.
- **Exit gate:** all V&V passed, residual risks acceptable, benefit-risk positive.

Phase F — Certification & market authorisation (months 16–24+)

Objective: obtain QMS certification and market clearance.

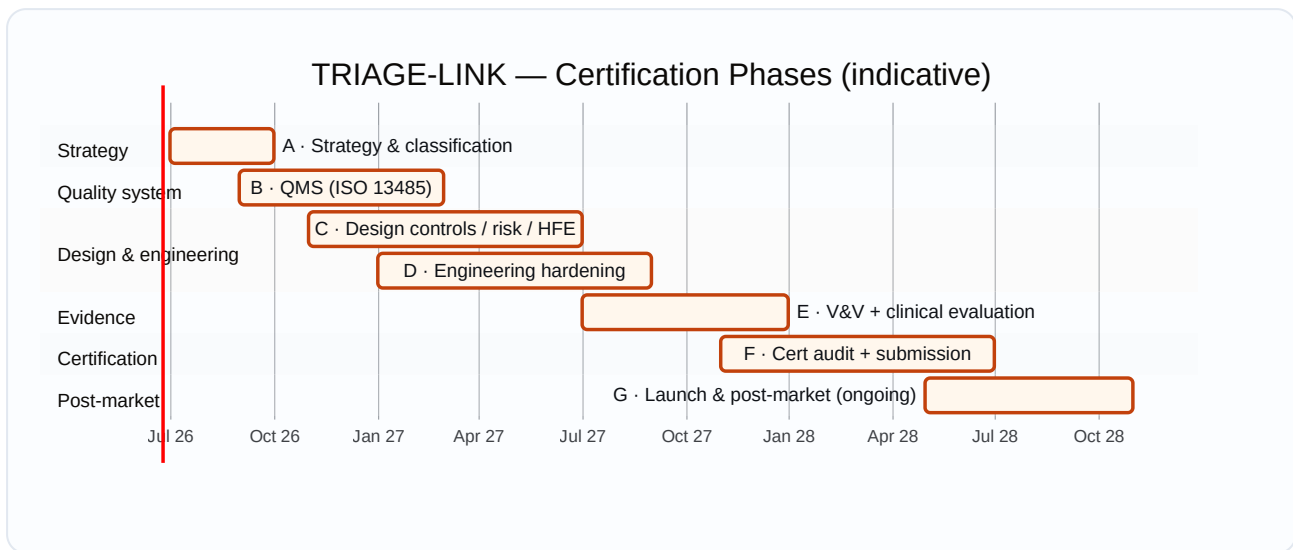
- **ISO 13485 certification audit** (Stage 1 + Stage 2) via registrar; **MDSAP** if multi-market.
- Compile the **Technical Documentation / submission:**
 - Canada: **MDL** application (Class II+ requires ISO 13485/MDSAP cert).
 - USA: **510(k)** (with predicate) or **De Novo**.
 - EU: **Technical Documentation** + Notified Body conformity assessment → **CE**.
- Respond to deficiency letters / audit findings; close nonconformities.
- **Deliverables:** ISO 13485 certificate, market authorisation(s), approved labelling/IFU, UDI.
- **Exit gate:** cleared/licensed/CE-marked for the launch market.

Phase G — Launch & post-market (ongoing)

Objective: sustain compliance in the field.

- **Post-Market Surveillance** plan; **vigilance** (incident/recall reporting per jurisdiction).
- **Complaint handling**, CAPA, periodic safety reporting (**PSUR/MDR**), trending.
- **Change management:** software changes assessed for re-submission/re-validation impact (significant-change determination); cybersecurity patch process.
- Surveillance audits (annual ISO 13485 / MDSAP); standards re-certification.
- **Deliverables:** PMS reports, vigilance records, CAPA log, change-control records.
- **Exit gate (continuous):** certification maintained; obligations met.

■ 7. Phase timeline



Indicative only — **18–30 months** prototype → cleared for a Class II SaMD is typical; EU MDR Notified Body capacity and any clinical investigation are the main schedule risks.

■ 8. Documentation & evidence artifacts

The **Technical Documentation / Device Master Record** a submission/audit expects:

- Intended Use / Indications for Use; classification rationale
- Device description & specifications; labelling and **Instructions For Use**; **UDI**
- **Design History File** (design inputs, outputs, reviews, transfer)
- **Risk Management File** (ISO 14971): plan, hazard analysis, risk controls, benefit-risk, report
- **Software documentation** (IEC 62304): dev plan, safety classification, architecture, detailed design, **SOUP list**, **SBOM**, V&V, anomaly list, release notes
- **Usability Engineering File** (IEC 62366): use-spec, use-related risk, formative + summative reports
- **Cybersecurity file** (IEC 81001-5-1): threat model, security risk assessment, pen-test, vuln-management plan
- **Verification & Validation** reports + **traceability matrix**
- **Clinical Evaluation** Plan/Report (+ clinical investigation records if applicable)
- **QMS records**: management review, internal audits, CAPA, supplier/SOUP control, training
- **Post-Market Surveillance** & vigilance plan; PSUR
- **Declaration of Conformity** (EU) / submission dossiers per market

■ 9. Organisation & roles

Role	Responsibility
Management Representative (ISO 13485)	QMS authority, management review
Regulatory Affairs	Strategy, classification, submissions, liaison with authorities/NB
Quality Manager	SOPs, audits, CAPA, document/record control
Software Lead / Architect	IEC 62304 lifecycle, architecture, SOUP/SBOM
Risk Manager	ISO 14971 file, hazard analysis, benefit-risk
Human Factors / Usability	IEC 62366 use-spec, formative/summative studies
Security Lead	IEC 81001-5-1, threat model, pen-test, vuln management
Clinical Lead	Clinical evaluation, intended-use clinical input, (investigation)
Privacy Officer	PHIPA/PIPEDA/HIPAA/GDPR compliance, DPIA

Small teams combine roles, but **independence of review** (e.g. tester ≠ author for V&V sign-off) must be preserved.

■ 10. Quality / decision gates

Gate	After	Criteria to pass
G0 — Go/No-Go	Phase A	Intended use, classification, pathway, safety class agreed; budget approved
G1 — QMS ready	Phase B	SOPs effective; tools validated; internal audit clean
G2 — Design frozen	Phase C	Inputs ↔ outputs traced; risk file baselined; design reviews passed
G3 — Security verified	Phase D	Threat model closed; pen-test high-severity findings remediated
G4 — V&V complete	Phase E	All tests traced & passed; summative usability passed; benefit-risk positive
G5 — Certified	Phase F	ISO 13485 cert issued; market authorisation granted
G6 — Surveillance	Phase G (recurring)	PMS active; CAPA closed in SLA; audits maintained

■ 11. Mapping: engineering roadmap → certification workstreams

The existing `docs/ROADMAP.md` build phases supply the technical substance; this roadmap puts them under design control and adds the regulatory/clinical evidence.

docs/ROADMAP.md (build)	Certification phase	Becomes evidence as...
Phase 0 — Gating decisions	A Strategy & classification	Intended Use + classification rationale
Phase 1 — Harden & extract core	C Design controls	Architecture design outputs + unit V&V
Phase 2 — Sync backend	C/E Design + V&V	Requirements, design, integration V&V, data-integrity risk controls
Phase 3 — Security	D Engineering hardening	Security risk assessment, controls, pen-test
Phase 4 — Hospital handover	C/E Design + V&V	Interoperability requirements + FHIR conformance evidence
Phase 5 — Deploy for real	D/G Hardening + PMS	Reliability/DR validation; production monitoring feeds vigilance
Anatomical body chart	C/E Design + V&V	Requirement + verification (and risk if clinically load-bearing)

■ 12. Indicative cost & timeline

Rough, market- and scope-dependent — for planning only, **confirm with a regulatory consultant**.

Workstream	Indicative effort/cost (Class II)
Regulatory strategy & consulting	\$30–80k
ISO 13485 QMS setup + certification	\$40–120k (setup) + annual surveillance
Risk, usability (HFE), security, pen-test	\$60–150k
V&V formalisation + clinical evaluation	\$50–150k (more with a clinical investigation)
Submission fees (per market)	Canada MDL low; FDA 510(k) user fee ~\$ tens of k; EU NB fees significant
Total, prototype → first market (Class II)	~\$250–700k and 18–30 months, typical

Class III, EU MDR, or a clinical investigation can multiply both. Sequencing markets (Canada → US → EU) spreads cost and de-risks.

■ 13. Key risks & mitigations

Risk	Impact	Mitigation
Mis-scoped intended use inflates class	Cost/time blow-up	Tighten claims in Phase A; consider documentation-tool framing for v1
Retrofitting evidence onto existing code	Rework, audit findings	Bring engineering under QMS early (Phases B–D in parallel)
EU MDR Notified Body capacity	Schedule slip	Launch Canada/US first; book NB slot early
Clinical evidence insufficient	Submission delay	Clinical Evaluation Plan in Phase A; literature + real-world evidence; investigation only if needed
Cybersecurity findings late	Re-validation	Threat model + pen-test in Phase D, not at the end
Usability validation fails summative	Redesign late	Formative studies throughout Phase C; pre-defined pass criteria
Offline/sync data-integrity hazards	Patient-safety risk	Deterministic merge + audit already designed; formalise as ISO 14971 risk controls
Scope creep across markets	Diffuse effort	Single launch market + MDSAP to amortise the QMS audit

■ 14. Glossary

Term	Meaning
SaMD	Software as a Medical Device — software with a medical purpose, not part of a hardware device
IMDRF	International Medical Device Regulators Forum — the SaMD risk-categorisation framework
ISO 13485	Quality Management System standard for medical devices
IEC 62304	Medical device software lifecycle standard (safety classes A/B/C)
ISO 14971	Medical device risk management standard
IEC 62366-1	Usability engineering / human factors standard
IEC 81001-5-1	Health software cybersecurity lifecycle standard
DHF	Design History File — the record of design controls
SOUP	Software Of Unknown Provenance — third-party components managed under IEC 62304
SBOM	Software Bill of Materials
HFE	Human Factors Engineering
V&V	Verification & Validation
MDSAP	Medical Device Single Audit Program (CA/US/AU/BR/JP)
MDL	Medical Device Licence (Health Canada)
510(k) / De Novo	FDA premarket pathways (predicate-based / novel low-moderate risk)
MDR	EU Medical Device Regulation 2017/745
Notified Body	EU third-party conformity-assessment organisation
PMS / PSUR	Post-Market Surveillance / Periodic Safety Update Report
UDI	Unique Device Identification
PHIPA / PIPEDA / HIPAA / GDPR	Health-data privacy laws (Ontario / Canada / US / EU)
CAPA	Corrective And Preventive Action

Companion: [docs/ROADMAP.md](#) (engineering build plan), [docs/MASTER-ARCHITECTURE.md](#) (system architecture incl. §10 Security and §11 Regulatory & compliance).